The Role of Artificial Intelligence in Transforming Secondary Education: Insights from Teachers in Rural, Semi-Urban, and Urban West Bengal

Abstract

The integration of Artificial Intelligence (AI) into education is poised to revolutionize teaching-learning processes by enhancing personalization, engagement, and efficiency. This study explores the awareness, challenges, and expectations surrounding AI adoption among secondary school teachers in seven districts of West Bengal, India, encompassing Hooghly, Purba Medinipur, Purulia, Jhargram, North 24 Parganas, Nadia, and Purba Bardhaman. Employing a descriptive survey research design, data were collected from 270 teachers across 54 schools, representing rural (83%), semi-urban (7%), and urban (10%) locations. The objectives of the study include examining ethical and privacy concerns, assessing teachers' perceptions of AI's potential to enhance educational outcomes, and evaluating professional development requirements for AI integration.Structured questionnaires were used to collect primary data. Findings reveal diverse perspectives on AI's role, with a significant interest in workshops focused on ethical implications (31%) and practical applications in lesson planning (28%). However, technical skill gaps and reluctance to participate in AI-focused training programmes emerged as critical barriers. Teachers expressed mixed opinions about AI's impact on

student engagement and personalization of learning, with 42% acknowledging its potential to enhance motivation and adaptability, while 30% showed indifference, highlighting the need for improved tools and awareness initiatives. Rural schools face unique challenges, such as limited infrastructure, further emphasizing the importance of tailored strategies for equitable AI adoption. The study concludes that successful integration of AI in secondary education requires a holistic approach, combining targeted professional development, ethical literacy, infrastructure improvements, and awareness campaigns. Policymakers and educators must collaborate to address disparities, build trust, and demonstrate AI's benefits to ensure its effective and inclusive implementation. This research provides valuable insights into the opportunities and challenges of AI in education, offering a roadmap for future initiatives aimed at fostering its transformative potential.